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VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

K. J. Somaiya College of Engineering, Mumbai-77

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Department of Electronics Engineering

Course Name:	Object Oriented Programming	Semester:	III
Date of Performance:	___ / ___ / ____	Batch No:	Batch _A4_
Faculty Name:		Roll No:	16015024082
Faculty Sign & Date:		Grade/Marks:	___/15

Experiment No: 4

Title: Concept of Inheritance and Abstract class

Aim and Objective of the Experiment:

Learn the concept of Inheritance, method overriding and Abstract class in Java

COs to be achieved:

CO2: Understand concepts of Object Oriented Programming and basic characteristics of Java.

Tools used:



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Theory:

1. (About Types of Inheritance and abstract class)

Inheritance in Java allows code reusability through single, multilevel, hierarchical, and interface-based multiple inheritance. Abstract classes provide a base with unimplemented methods, letting subclasses define specific behaviors.

Code:

1. Create a class named '*Rectangle*' with two data members '*length*' and '*breadth*' and two methods to print the area and perimeter of the rectangle respectively. Its constructor having parameters for length and breadth is used to initialize length and breadth of the rectangle. Let class '*Square*' inherit the '*Rectangle*' class with its constructor having a parameter for its side (suppose s) calling the constructor of its parent class as '*super(s,s)*'. Print the area and perimeter of a rectangle and a square.
2. Now repeat the above example to print the area of 10 squares. Hint-Use array of objects
3. Write a java program to create an abstract class named Shape that contains two integers and an empty method named print Area (). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method print Area () that prints the area of the given shape.



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```
1 package Randoms;
2
3 class Rectangle { 3 usages new *
4     int length, breadth; 3 usages
5
6     Rectangle(int length, int breadth) { 2 usages new *
7         this.length = length;
8         this.breadth = breadth;
9     }
10
11     void printArea() { 2 usages new *
12         int area = length * breadth;
13         System.out.println("Area of Rectangle = " + area);
14     }
15
16     void printPerimeter() { 2 usages new *
17         int perimeter = 2 * (length + breadth);
18         System.out.println("Perimeter of Rectangle = " + perimeter);
19     }
20 }
21
22 class Square extends Rectangle { 2 usages new *
23
24     Square(int side) { super(side, side); }
25 }
26
27
28
29 public class Main { 1 utakarsh23 *
30     public static void main(String[] args) { new *
31         Rectangle rect = new Rectangle( length: 10, breadth: 5);
32         rect.printArea();
33         rect.printPerimeter();
34
35         Square sq = new Square( side: 7);
36         sq.printArea();
37         sq.printPerimeter();
38     }
39 }
```

1.



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2.

```
1 package Randoms;
2
3 class Rectangle { new *
4     int length, breadth;
5
6     Rectangle(int length, int breadth) { new *
7         this.length = length;
8         this.breadth = breadth;
9     }
10
11     void printArea() { new *
12         System.out.println("Area = " + (length * breadth));
13     }
14
15     void printPerimeter() { new *
16         System.out.println("Perimeter = " + (2 * (length + breadth)));
17     }
18 }
19
20 class Square extends Rectangle { new *
21     Square(int side) { new *
22         super(side, side);
23     }
24 }
25
26 public class Main { utakarsh23 *
27     public static void main(String[] args) { new *
28         Square[] squares = new Square[10];
29
30         for (int i = 0; i < squares.length; i++) {
31             squares[i] = new Square(side: i + 1);
32         }
33
34         for (int i = 0; i < squares.length; i++) {
35             System.out.print("Square " + (i + 1) + " → ");
36             squares[i].printArea();
37         }
38     }
39 }
```



```
3  @ abstract class Shape { 6 usages 3 inheritors new *
4      int dim1, dim2; 5 usages
5      Shape(int d1, int d2) { 3 usages new *
6          dim1 = d1;
7          dim2 = d2;
8      }
9  @ abstract void printArea(); 3 usages 3 implementations new *
10 }
11 class Rectangle extends Shape { 1 usage new *
12     Rectangle(int length, int breadth) { 1 usage new *
13         super(length, breadth);
14     }
15  @ void printArea() { 3 usages new *
16     System.out.println("Area of Rectangle = " + (dim1 * dim2));
17 }
18 }
19 class Triangle extends Shape { 1 usage new *
20     Triangle(int base, int height) { 1 usage new *
21         super(base, height);
22     }
23  @ void printArea() { 3 usages new *
24     System.out.println("Area of Triangle = " + (0.5 * dim1 * dim2));
25 }
26 }
27 class Circle extends Shape { 1 usage new *
28     Circle(int radius) { 1 usage new *
29         super(radius, 0);
30     }
31  @ void printArea() { 3 usages new *
32     System.out.println("Area of Circle = " + (Math.PI * dim1 * dim1));}}
33  > public class Main { 1 utakarsh23 *
34  >     public static void main(String[] args) { new *
35         Shape r = new Rectangle( length: 10, breadth: 5);
36         r.printArea();
37         Shape t = new Triangle( base: 10, height: 8);
38         t.printArea();
39         Shape c = new Circle( radius: 7);
40         c.printArea();
41     }
```

3.



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Output:

1.

```
spath /Users/utkarshmani/Desktop/DSA/target/classes Randoms.Main  
  
Area of Rectangle = 50  
Perimeter of Rectangle = 30  
Area of Rectangle = 49  
Perimeter of Rectangle = 28  
  
Process finished with exit code 0
```

2

```
top/DSA/target/classes Randoms.Main  
  
Square 1 → Area = 1  
Square 2 → Area = 4  
Square 3 → Area = 9  
Square 4 → Area = 16  
Square 5 → Area = 25  
Square 6 → Area = 36  
Square 7 → Area = 49  
Square 8 → Area = 64  
Square 9 → Area = 81  
Square 10 → Area = 100  
  
Process finished with exit code 0
```



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3

```
ni/Desktop/DSA/target/classes Randoms.Main
```

```
Area of Rectangle = 50
```

```
Area of Triangle = 40.0
```

```
Area of Circle = 153.93804002589985
```

```
Process finished with exit code 0
```

Post Lab Subjective/Objective type Questions:

1. Explain *super* keyword with all its usages. Support explanation with a program.
-The *super* keyword is used in inheritance. It ensures proper constructor chaining, it resolves variable/method hiding, and helps reuse parent functionality.



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```
1 package Randoms;
2
3 class Parent { 1 usage new *
4     String name = "Parent"; 1 usage
5
6     Parent() { 1 usage new *
7         System.out.println("Parent constructor called");
8     }
9
10    void showMessage() { 1 usage new *
11        System.out.println("Hello from Parent class");
12    }
13 }
14
15 class Child extends Parent { 2 usages new *
16     String name = "Child"; 1 usage
17
18     Child() { 1 usage new *
19         super();
20         System.out.println("Child constructor called");
21     }
22
23     void display() { 1 usage new *
24         System.out.println("Child name = " + name);
25
26         System.out.println("Parent name = " + super.name);
27
28         super.showMessage();
29     }
30 }
31
32 public class Main { 1 utakarsh23 *
33     public static void main(String[] args) { new *
34         Child c = new Child();
35         c.display();
36     }
37 }
```



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1. Perform following virtual lab experiments.

- a. <https://java-iitd.vlabs.ac.in/exp/inheritance/>
- a. <https://java-iitd.vlabs.ac.in/exp/method-overloading/>
- a. <https://java-iitd.vlabs.ac.in/exp/method-overriding/>



Implementing Overriding through Polymorphism



Rate Me

Report a Bug

Preview

Simulator

execute

Output

I am happy

Syntax

```
public class ClassName{
    public void FunctionName()
    {
        System.out.println("text1");
    }
}
public class ClassName extends BaseClassName{
    public void FunctionName()
    {
        System.out.println("text2");
    }
}
```



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Implementing Method Overloading through Polymorphism

☆☆☆☆☆

Rate Me

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Preview

Simulator

execute

Output

29
60

Syntax

public class ClassName {
Returntype MethodName (datatype1 variable1, datatype2 variable2)
{
System.out.println(variable1+variable2);
}
Returntype MethodName (datatype1 variable1, datatype2 variable2,
datatype3 variable3){
System.out.println(variable1+variable2+variable3);
}

Understanding Inheritance in Java

☆☆☆☆☆

Rate Me

Report a Bug

Preview

Simulator

execute

Output

12
13

Syntax

public class ClassName {
int VariableName = VariableValue ;
}
public class SubClassName extends SuperClassName{
int VariableName = VariableValue ;
public static void main(String[] args){
SubClassName obj = new SubClassName() ;
System.out.println(ObjectName.SuperClassVariable) ;
}

Conclusion:



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In this work, we explored key Java concepts like inheritance, abstraction, and the use of super, showing how these ideas simplify coding, improve reusability, and make programs more organized and easier to understand.

Signature of faculty in-charge with Date: